

Daily Geometry Drill #2

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Circle Equations (centre $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$) · Tangents · Line–Circle Tests · Chord Lengths.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Complete the square, read the centre and radius, then write or test circle data on sight.

- Write down the centre of the circle $x^2 + y^2 - 6x - 8y + 24 = 0$.
- For which value of k does $x^2 + y^2 - 6x + 4y + k = 0$ represent a circle of radius 5?
- Write down the equation of the circle with centre $(2, -1)$ and radius 5.
- Write down the radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$.
- The point $(x, 4)$ lies on the circle $x^2 + y^2 = 25$ with $x > 0$. Find x .
- The radius from the origin to the point $(3, 4)$ on $x^2 + y^2 = 25$ has what gradient?
- The tangent to $x^2 + y^2 = 25$ at the point $(3, 4)$ has what gradient?
- Write down the equation of the circle with centre $(3, -4)$ that passes through the origin.
- The centre of $x^2 + y^2 - 6x - 8y + 24 = 0$ is $(3, 4)$. Find the perpendicular distance of this centre from the line $4x + 3y - 9 = 0$.
- A chord of the circle $x^2 + y^2 = 25$ is at perpendicular distance 3 from the origin. Find the length of the chord.

Section B Manipulation Drills

(≤50 s each)

Combine the new circle tools with last drill's line machinery. Complete the square first, then test.

- Which of the following is the equation of the circle with centre $(3, -2)$ and radius 4?
 - $x^2 + y^2 - 6x + 4y - 3 = 0$
 - $x^2 + y^2 - 6x + 4y + 9 = 0$
 - $x^2 + y^2 + 6x - 4y - 3 = 0$

- D) $x^2 + y^2 - 6x - 4y - 3 = 0$
2. The circle $x^2 + y^2 + 2x - 4y - 11 = 0$ has what radius?
- A) 2
B) 3
C) 4
D) $\sqrt{5}$
3. The point $(1, 2)$ lies on the circle $x^2 + y^2 = 5$. The equation of the tangent to the circle at $(1, 2)$ is?
- A) $y = -\frac{1}{2}x + \frac{5}{2}$
B) $y = 2x$
C) $y = -2x + 5$
D) $y = \frac{1}{2}x$
4. A circle of form $x^2 + y^2 + 2gx + 2fy + c = 0$ passes through the origin. Deduce c .
- A) $c = 1$
B) $c = 0$
C) $c = g + f$
D) $c = -1$
5. The perpendicular distance from the centre of $x^2 + y^2 = 25$ to the line $y = 2x + 3$ is?
- A) $\frac{3}{\sqrt{5}}$
B) $\frac{3}{5}$
C) 3
D) 2
6. Which of the following is a tangent to the circle $x^2 + y^2 = 4$?
- A) $y = x + 2\sqrt{2}$
B) $y = x + 4$
C) $y = 2x + 2$
D) $y = 2x + 4$
7. A chord of the circle $x^2 + y^2 = 13$ has length $4\sqrt{3}$. What is the perpendicular distance of the chord from the centre?
- A) 3
B) 2

- C) $\sqrt{5}$
D) 1
8. The line $y = mx + 4$ is a tangent to the circle $x^2 + y^2 = 4$. Find the two possible values of m .
A) ± 1
B) $\pm\sqrt{3}$
C) ± 2
D) $\pm\sqrt{2}$
9. The circle with centre $(6, a)$ and radius 6 is tangent to the y -axis and passes through the point $(0, 8)$. Find a .
A) 1
B) 3
C) 5
D) 8
10. Write down the length of the chord cut from the line $y = x + 1$ by the circle $x^2 + y^2 = 25$.

Section C Substitution & Structure(≤ 90 s each)

Hard: hidden tangency, non-obvious chord bisection, and the distance-vs-discriminant choice. No brute substitution.

1. A tangent to the circle $x^2 + y^2 = 144$ passes through $(20, 0)$ and meets the positive y -axis at T . What is the y -coordinate of T ? (*after TMUA 2017 Paper 1 Q6*)
A) 12
B) 15
C) $\frac{49}{3}$
D) 20
2. The point $P(8, 6)$ is outside the circle $x^2 + y^2 = 36$. Two tangents from P touch the circle at A and B . What is the length AB (the common chord of contact)? (*adapted from TMUA 2018 Paper 1 Q7*)
A) $\frac{12\sqrt{13}}{13}$
B) $\frac{24\sqrt{13}}{13}$
C) $\frac{48}{5}$
D) $\frac{72}{5}$
3. The point Q moves on the circle $x^2 + y^2 - 2x + 4y - 20 = 0$, and P is the point $(13, 3)$. What is the least possible distance PQ ?
A) 13

- B) 18
C) 12
D) 8
4. A circle has equation $x^2 + y^2 - 18x - 22y + 178 = 0$. A regular hexagon $ABCDEF$ is inscribed so that all its vertices lie on the circle. What is the area of the hexagon? (*adapted from TMUA 2017 Paper 1 Q9*)
A) $18\sqrt{3}$
B) $36\sqrt{3}$
C) $\frac{75\sqrt{3}}{2}$
D) 75
5. For which values of p does $x^2 - 2px + y^2 - 6y - p^2 + 8p + 9 = 0$ represent a (real) circle? (*after TMUA 2022 Paper 1 Q2*)
A) $p < 0$ or $p > 4$
B) $-1 < p < 9$
C) $0 < p < 4$
D) $p < -1$ or $p > 9$
6. The circle $x^2 + y^2 - 4x + 2y - 11 = 0$ and the line $x - y + 1 = 0$: determine the number of intersection points without solving the system. (*after MAT 2019 Q6*)
A) 0 (line misses)
B) 1 (tangent)
C) 2 (secant)
D) It cannot be decided without solving the system
7. The circle $x^2 + y^2 - 8x - 4y - 21 = 0$ is cut by the y -axis. What is the length of the chord intercepted on the y -axis?
A) $2\sqrt{21}$
B) $2\sqrt{41}$
C) 8
D) 10
8. The line $y = kx + 5$ is tangent to the circle $x^2 + y^2 - 6x - 8y + 24 = 0$. How many distinct values of k satisfy this? (*after TMUA 2020 Paper 1 Q7*)
A) 0
B) 1
C) 2
D) Infinitely many

Section D Challenge

(TMUA / SMC / BMO1 difficulty)

Hardest 5: distance invariants, hidden right angles, and a construction you must spot.

1. The circles $C_1 : (x + 2)^2 + (y - 3)^2 = 18$ and $C_2 : (x - 7)^2 + (y + 6)^2 = 2$ have centres O_1, O_2 . What is the *shortest* distance between a point on C_1 and a point on C_2 ? (*adapted from TMUA 2018 Paper 1 Q7 / 2022 Paper 1 Q3*)
 - A) $5\sqrt{2} - 4$
 - B) $5\sqrt{2} - 5$
 - C) $5\sqrt{2}$
 - D) $5\sqrt{2} + 5$
2. The segment joining $(3, 3)$ and $(7, 5)$ is a diameter of circle C . C is translated 3 left, reflected in the x -axis, then enlarged scale factor 4 about its centre. What is the equation of the final circle? (*after TMUA Specimen Paper 1 Q9*)
 - A) $(x - 2)^2 + (y - 4)^2 = 80$
 - B) $(x - 2)^2 + (y + 4)^2 = 80$
 - C) $(x - 2)^2 + (y - 4)^2 = 320$
 - D) $(x - 2)^2 + (y + 4)^2 = 320$
3. A circle has centre O and radius 6. Points P and Q lie on it with $\angle POQ \geq 90^\circ$, and triangle POQ has area $9\sqrt{3}$. The point R moves on the major arc PQ . What is the greatest possible area of triangle PRQ ? (*adapted from TMUA 2022 Paper 1 Q14*)
 - A) $18 + 9\sqrt{3}$
 - B) $27\sqrt{3}$
 - C) $27 + 9\sqrt{3}$
 - D) $36 + 9\sqrt{3}$
4. Two tangents are drawn from the origin to the circle $(x - 5)^2 + (y - 5)^2 = 5$. What is the tangent of the acute angle between them?
 - A) $\frac{3}{4}$
 - B) $\frac{4}{3}$
 - C) $\frac{5}{2}$
 - D) 1
5. The circle through $(0, 0)$, $(8, 0)$ and $(2, 6)$ meets the y -axis again at Q . How far is Q from the origin?
 - A) $2\sqrt{5}$
 - B) 6
 - C) 2
 - D) 4

“Tangency is a distance statement in disguise: compare d to r , never solve the intersection.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths